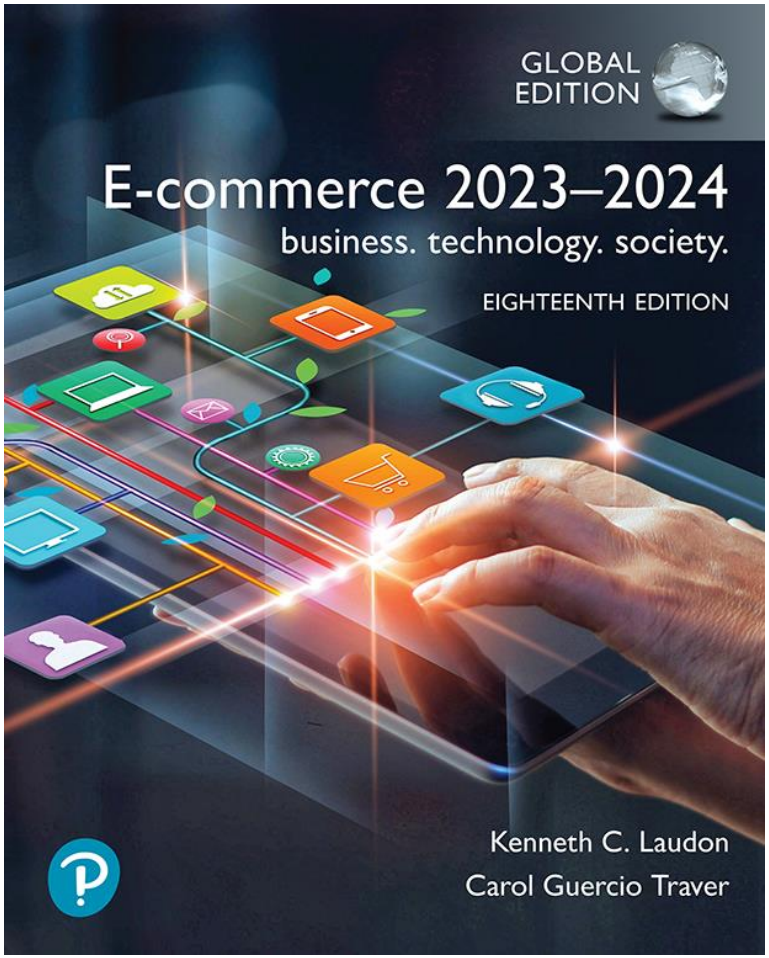
# E-commerce 2023–2024: business. technology. society.

Eighteenth Edition




## Chapter 4

Building an E-commerce Presence

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# Learning Objectives

**4.1** Understand the questions you must ask and answer, and the steps you should take, in developing an e-commerce presence.

**4.2** Explain the process that should be followed in building an e-commerce presence.

**4.3** Identify and understand the major considerations involved in choosing web server and e-commerce merchant server software.

**4.4** Understand the issues involved in choosing the most appropriate hardware for an e-commerce site.

**4.5** Identify additional tools that can improve website performance.

**4.6** Understand the important considerations involved in developing a mobile website and building mobile applications.

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# Scratch Builds an Online Presence

- Class Discussion
  - What were Scratch's e-commerce objectives?
  - What were the conditions that made it ideal opportunity for Scratch to establish an e-commerce presence?
  - How has Scratch adapted the way it works to cater to the demands of mobile device users?

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# Imagine Your E-commerce Presence (1 of 3)

- What's the idea? The vision includes:
  - Mission statement
  - Target audience
  - Intended market space
  - Strategic analysis
  - Marketing matrix
  - Development timeline
  - Preliminary budget

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# Imagine Your E-commerce Presence (2 of 3)

- Where's the money?
  - Business model(s)
  - Revenue model(s)
- Who and where is the target audience?
  - Demographics, lifestyle, consumption patterns, etc.
- What is the ballpark? Characterize the marketplace
  - Size, growth, demographics, structure

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# Imagine Your E-commerce Presence (3 of 3)

- Where's the content coming from?
- Know yourself: SWOT analysis
- Develop an e-commerce presence map
- Develop a timeline: Milestones
- How much will this cost?
  - Simple website: up to $5000
  - Small startup: $25,000 to $50,000
  - Large corporate website: $100,000+ to millions

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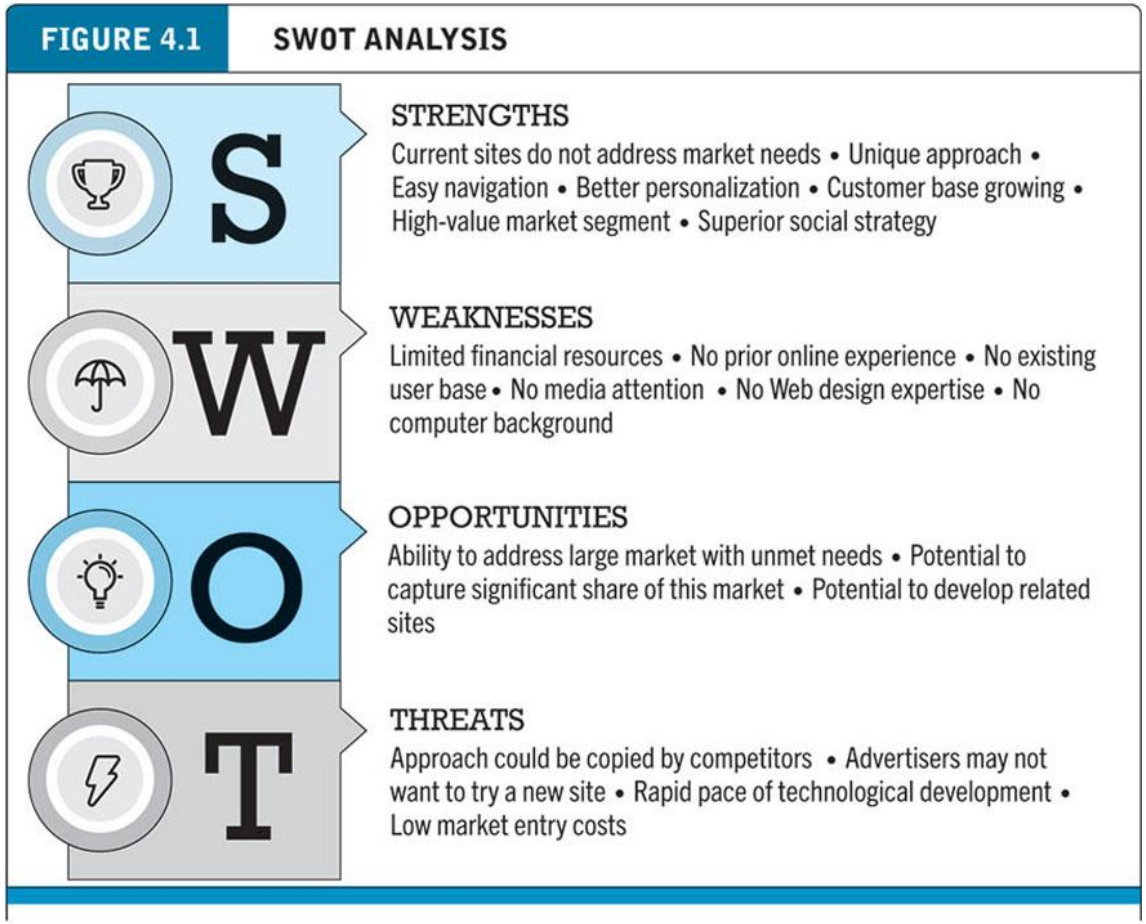
# Figure 4.1 SWOT Analysis

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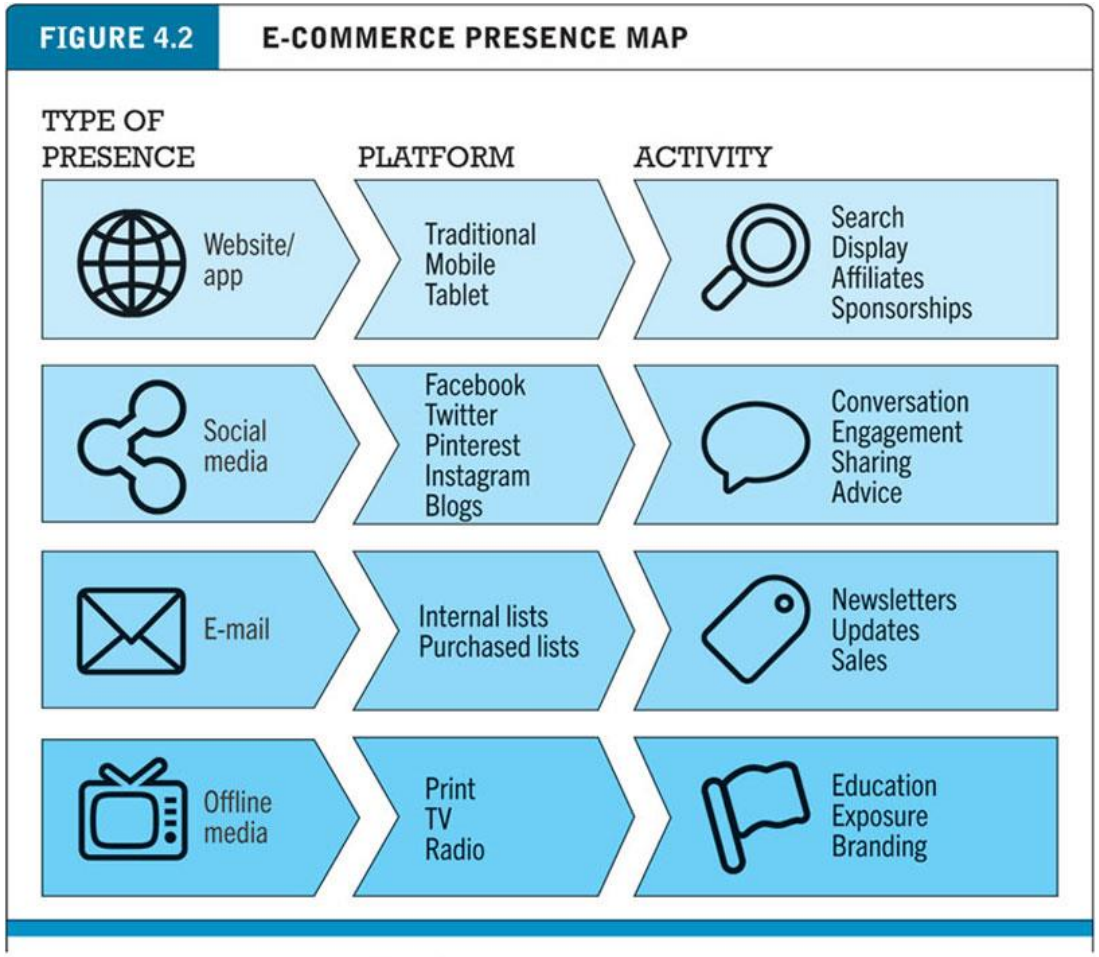
# Figure 4.2 E-commerce Presence Map

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# Building an E-commerce Site: A Systematic Approach

- Most important management challenges:
  1. Developing a clear understanding of business objectives
  2. Knowing how to choose the right technology to achieve those objectives
- Main factors to consider
  - Management
  - Hardware architecture
  - Software
  - Design
  - Telecommunications
  - Human resources

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# Planning: The Systems Development Life Cycle

- Methodology for understanding business objectives of a system and designing an appropriate solution
- Five major steps
  - Systems analysis/planning
  - Systems design
  - Building the system
  - Testing
  - Implementation

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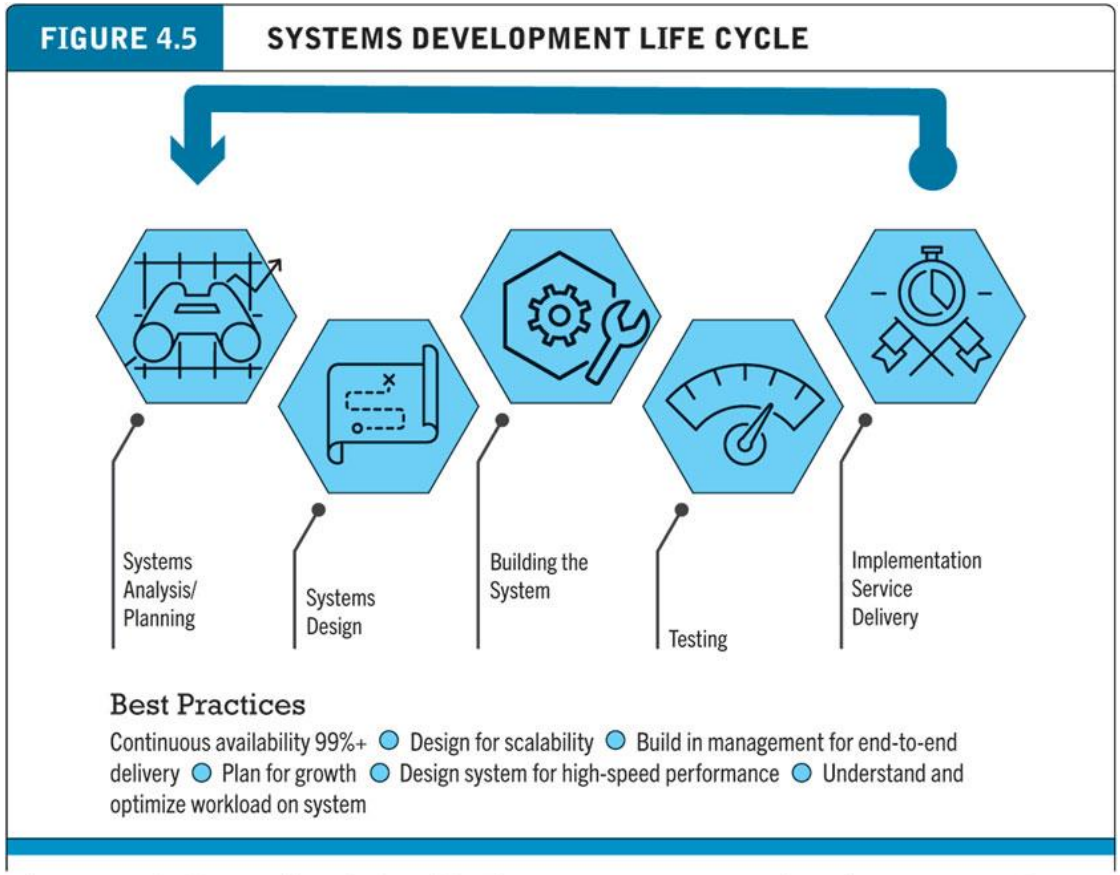
# Figure 4.5 Systems Development Life Cycle




The systems development life cycle (SDLC) has five major steps: systems analysis/planning, systems design, building the system, testing, and implementation and maintenance.

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# System Analysis/Planning

- Business objectives
  - List of capabilities you want your site to have
- System functionalities
  - List of information system capabilities needed to achieve business objectives
- Information requirements
  - Information elements that system must produce in order to achieve business objectives

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# Table 4.2 System Analysis, Business Objectives, System Functionalities, and Information Requirements for a Typical E-commerce Site (1 of 2)

| Business Objective | System Functionality | Information Requirements |
|---|---|---|
| Display goods | Digital catalog | Dynamic text and graphics catalog |
| Provide product information | Product database | Product description, stocking numbers, inventory levels |
| Personalize/customize product | Customer on-site tracking | Site log for every customer visit; data mining capability to identify common customer paths and appropriate responses |
| Engage customers in conversations | On-site blog; user forums | Customer comments captured by software with blogging and community forum functionality |
| Execute a transaction | Shopping cart/payment system | Secure credit card clearing; multiple payment options |
| Accumulate customer information | Customer database | Name, address, phone, and e-mail for all customers; online customer registration |

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# Table 4.2 System Analysis, Business Objectives, System Functionalities, and Information Requirements for a Typical E-commerce Site (2 of 2)

| Business Objective | System Functionality | Information Requirements |
|---|---|---|
| Provide after-sale customer support | Sales database | Customer ID, product, date, payment, shipment date |
| Coordinate marketing/advertising | Ad server, e-mail server, e-mail, campaign manager, ad banner manager | Site behavior log of prospects and customers linked to e-mail and banner ad campaigns |
| Understand marketing effectiveness | Site tracking and reporting system | Number of unique visitors, pages visited, products purchased, identified by marketing campaign |
| Provide production and supplier links | Inventory management system | Product and inventory levels, supplier ID and contact, order quantity data by product |

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# Systems Design: Hardware and Software Platforms

- System design specification
  - Description of main components of a system and their relationship to one another
- Two components of system design:
  - Logical design
    - Data flow diagrams, processing functions, databases
  - Physical design
    - Specifies actual physical, software components, models, and so on

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# Building the System: In-House Versus Outsourcing

- Outsourcing: Hiring vendors to provide services involved in building site
- Build own vs outsourcing
  - Build your own requires team with diverse skill set; choice of software tools; both risks and possible benefits
- Host own vs outsourcing
  - Hosting: Hosting company responsible for ensuring site is accessible 24/7, for monthly fee
  - Co-location: Firm purchases or leases web server (with control over its operation), but server is located at vendor's facility

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# Figure 4.7 Choices in Building and Hosting

**FIGURE 4.7 CHOICES IN BUILDING AND HOSTING**

| HOSTING THE SITE | BUILDING THE SITE: In-house | BUILDING THE SITE: Outsource |
|---|---|---|
| In-house | COMPLETELY IN-HOUSE<br>Build: In<br>Host: In | MIXED RESPONSIBILITY<br>Build: Out<br>Host: In |
| Outsource | MIXED RESPONSIBILITY<br>Build: In<br>Host: Out | COMPLETELY OUTSOURCED<br>Build: Out<br>Host: Out |

You have a number of alternatives to consider when building and hosting an e-commerce site.

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# Insight on Business: OVH Takes E-Commerce to the Clouds

- Class Discussion
  - What value does OVH offer to businesses?
  - Are there any drawbacks to using OVH to create an e-commerce presence?
  - How are service providers like OVH changing the nature of e-commerce?

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# Testing the System

- Testing
  - Unit testing
  - System testing
  - Acceptance testing
  - A/B testing (split testing)
  - Multivariate testing

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# Implementation, Maintenance, and Optimization

- Systems break down unpredictably
- Maintenance is ongoing
- Maintenance costs: Similar to development costs
  - A $40,000 e-commerce site may require $40,000 annually to upkeep
- Benchmarking

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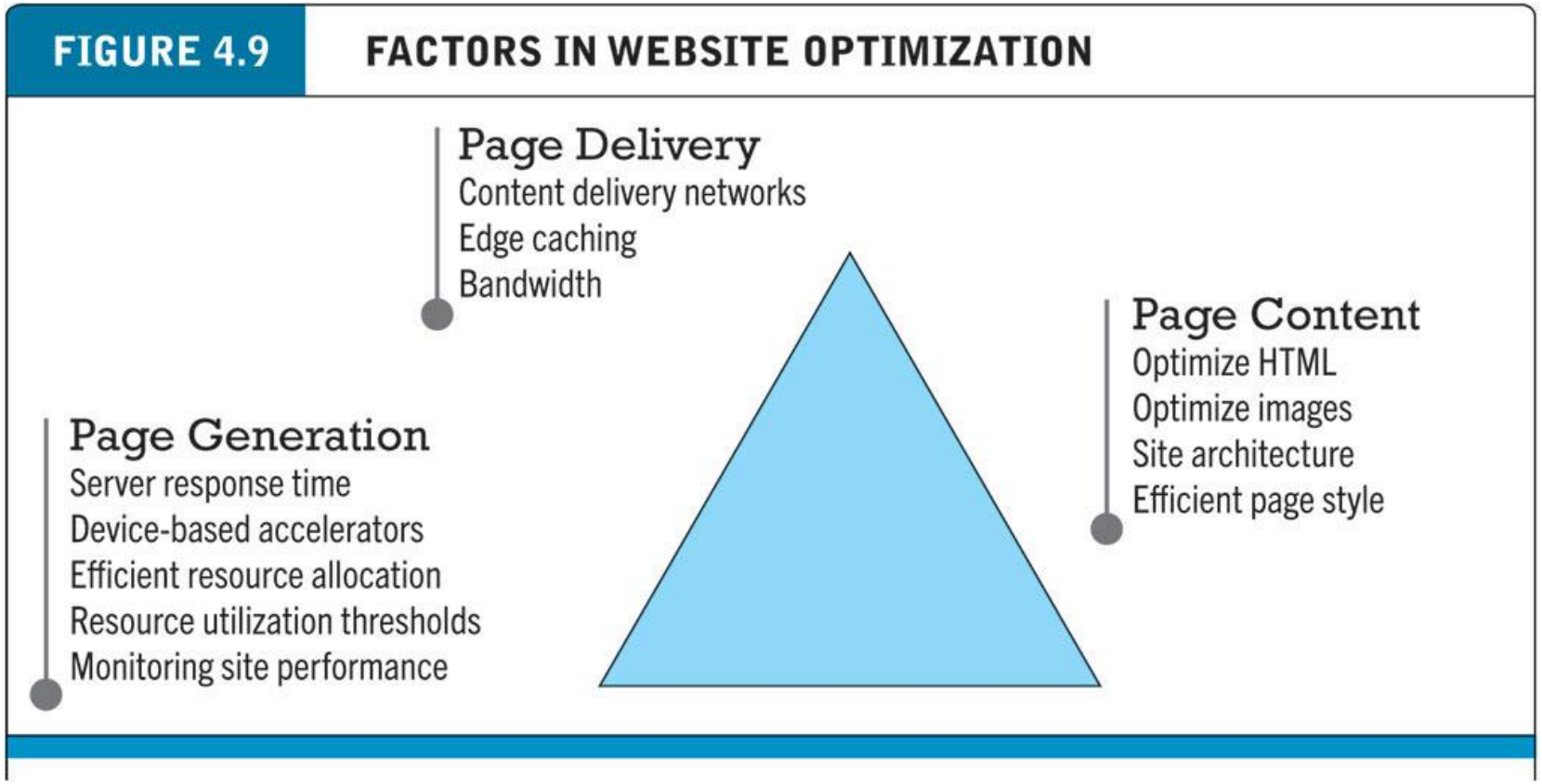
# Figure 4.9 Factors in Website Optimization

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# Alternative Web Development Methodologies

- Prototyping
- Agile development
- DevOps
- Component-based development
- Web services
  - Service-oriented architecture (SOA)
  - Microservices

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# Simple vs Multi-Tiered Website Architecture

- System architecture
  - Arrangement of software, machinery, and tasks in an information system needed to achieve a specific functionality
- Two-tier
  - Web server and database server
- Multi-tier
  - Web application servers
  - Backend, legacy databases

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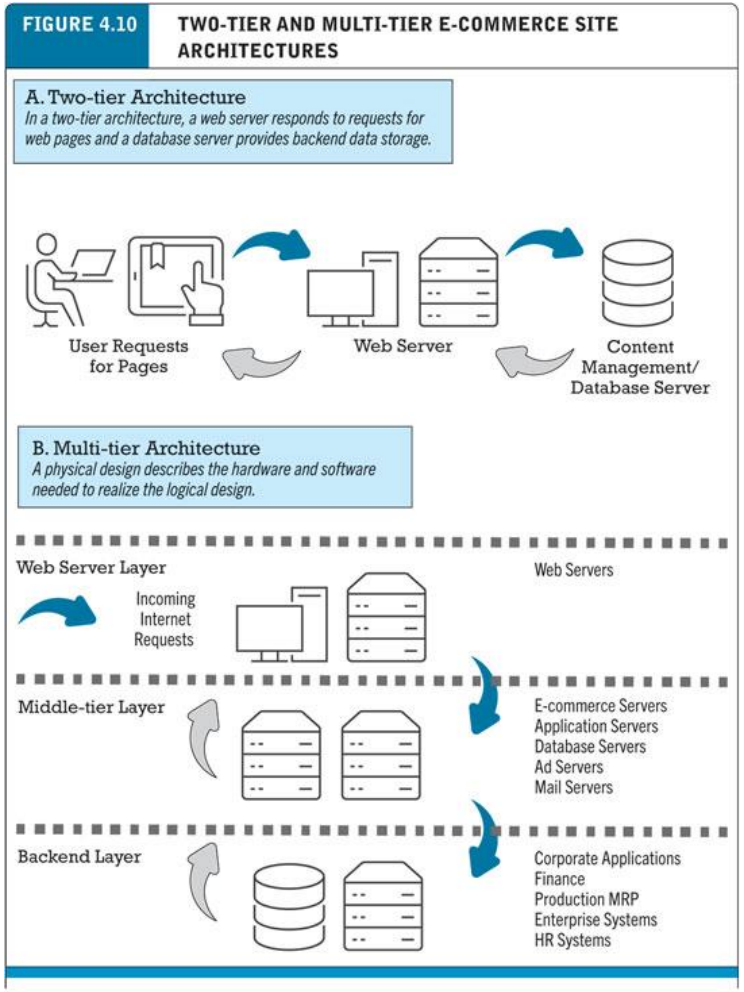
# Figure 4.10 Two-Tier and Multi-tier E-commerce Site Architectures

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# Web Server Software

- Apache
  - Leading web server software
  - Works with UNIX, Linux operating systems
  - Reliable, stable, part of open software community
- Microsoft's Internet Information Server (IIS)
  - Second major web server software
  - Windows-based
  - Integrated, easy-to-use

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# Site Management Tools

- Basic tools included in all web servers
  - Verify that links on pages are still valid
  - Identify orphan files
- Third-party software for advanced management
  - Monitor customer purchases
  - Marketing campaign effectiveness
  - Keep track of hit counts and other statistics
  - Example: Webtrends

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# Dynamic Page Generation Tools

- Dynamic HTML (DHTML): used to change a way a web page looks but does not generate a unique web page
- Dynamic page generation
  - Contents stored in database and fetched when needed to create a unique web page
- Advantages
  - Lowers menu costs
  - Permits easy online market segmentation
  - Enables cost-free price discrimination
  - Enables web content management system (WCMS)

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# Application Servers

- Web application servers
  - Provide specific business functionality required for a website
  - Type of middleware
    - Isolate business applications from Web servers and databases
  - Single-function applications being replaced by integrated software tools that combine all functionality needed for e-commerce site

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# E-commerce Merchant Server Software (E-commerce Software Platforms) (1 of 3)

- Provides basic functionality for sales
  - Online catalog
    - List of products available on website
  - Shopping cart
    - Allows shoppers to set aside, review, edit selections, and then make purchase
  - Credit card processing
    - Typically works in conjunction with shopping cart
    - Verifies card and puts through credit to company's account at checkout

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# E-commerce Merchant Server Software (E-commerce Software Platforms) (2 of 3)

- Different options for different-sized businesses
  - Small and medium-sized businesses: Shopify; WordPress (WooCommerce), Wix, Square, Weebly, Squarespace, Bigcommerce, open-source solutions
  - Mid-range: HCL Commerce; Sitecore Experience Commerce
  - High-end: SAP Commerce, Oracle ATG Web Commerce, Adobe Commerce
- Many now also available as cloud-based SaaS solutions.

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# E-commerce Merchant Server Software (E-commerce Software Platforms) (3 of 3)

- Key factors in selecting a package
  - Functionality
  - Support for different business models, including m-commerce
  - Business process modeling tools
  - Visual site management and reporting
  - Performance and scalability
  - Connectivity to existing business systems
  - Compliance with standards
  - Global and multicultural capability
  - Local sales tax and shipping rules

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# Choosing Hardware

- Hardware platform
  - Underlying computing equipment needed for e-commerce functionality
- Objective
  - Enough platform capacity to meet peak demand without wasting money
- Important to understand the factors that affect speed, capacity, and scalability of a site

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# Right-Sizing Your Hardware Platform: The Demand Side

- Customer demand
  - Most important factor affecting speed of site
- Factors in overall demand
  - Number of simultaneous users in peak periods
  - Nature of customer requests (user profile)
    - E.g., power user like day traders, browsers, mobile users
  - Type of content (dynamic vs static Web pages)
  - Required security
  - Number of items in inventory (database query VS RAM)
  - Number of page requests
  - Speed of legacy applications

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# Right-Sizing Your Hardware Platform: The Supply Side

- Scalability
  - Ability of site to increase in size as demand warrants
- Ways to scale hardware
  - Vertically
    - Increase processing power of individual components
  - Horizontally
    - Employ multiple computers to share workload
  - Improve processing architecture
  - Outsource hosting, use content delivery network

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# Table 4.6 Vertical and Horizontal Scaling Techniques

| Technique | Application |
|---|---|
| Use a faster computer. | Deploy edge servers, presentation servers, data servers, etc. |
| Create a cluster of computers. | Use computers in parallel to balance loads. |
| Use appliance servers. | Use special-purpose computers optimized for their task. |
| Segment workload. | Segment incoming work to specialized computers. |
| Batch requests. | Combine related requests for data into groups, process as group. |
| Manage connections. | Reduce connections between processes and computers to a minimum. |
| Aggregate user data. | Aggregate user data from legacy applications in single data pools. |
| Cache. | Store frequently used data in cache rather than on the disk. |

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# Table 4.7 Improving the Processing Architecture of Your Site

| Architecture Improvement | Description |
|---|---|
| Separate static content from dynamic content. | Use specialized servers for each type of workload. |
| Cache static content. | Increase RAM to the gigabyte range and store static content in RAM. |
| Cache database lookup tables. | Use cache tables used to look up database records. |
| Consolidate business logic on dedicated servers. | Put shopping cart, credit card processing, and other CPU-intensive activity on dedicated servers. |
| Optimize ASP code. | Examine your code to ensure it is operating efficiently. |
| Optimize the database schema. | Examine your database search times and take steps to reduce access times. |

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# Other E-commerce Site Tools

- Website design: Basic business considerations
  - Enabling customers to find and buy what they need
- Tools for search engine optimization
  - Search engine placement
    - Metatags, titles, content
    - Identify market niches
    - Offer expertise
    - Links
    - Buy ads
    - Local e-commerce

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# E-commerce Website Features That Annoy Customers (1 of 2)

- Requiring user to view ad or intro page before going to website content
- Pop-up and pop-under ads and windows
- Too many clicks to get to the content
- Links that don’t work
- Confusing navigation; no search function
- Requirement to register and log in before viewing content or ordering
- Slow loading pages
- Content that is out of date

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# E-commerce Website Features That Annoy Customers (2 of 2)

- Inability to use browser's Back button
- No contact information available (web form only)
- Unnecessary splash/flash screens, animation, etc.
- Music or other audio that plays automatically
- Unprofessional design elements
- Text not easily legible due to size, color, format
- Typographical errors
- No or unclear returns policy

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# Table 4.9 The Eight Most Important Factors in Successful E-commerce Site Design

| Factor | Description |
|---|---|
| Functionality | Pages that work, load quickly, and point the customer toward your product offerings |
| Informational | Links that customers can easily find to discover more about you and your products |
| Ease of use | Simple foolproof navigation |
| Redundant navigation | Alternative navigation to the same content |
| Ease of purchase | One or two clicks to purchase |
| Multi-browser functionality | Site works with the most popular browsers |
| Simple graphics | Avoids distracting, obnoxious graphics and sounds that the user cannot control |
| Legible text | Avoids backgrounds that distort text or make it illegible |

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# Tools for Interactivity and Active Content

- Java, JSP, and JavaScript (including Node.js)
  - JavaScript-based tools: React, Vue, Angular JS, D3, jQuery and Ajax)
  - Typescript
- ASP (Active Server Pages)/ASP.NET
- ColdFusion
- PHP, Ruby on Rails, Django
- Other design elements:
  - Widgets, mashups

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# Personalization Tools

- Personalization: ability to treat people based on personal qualities and prior history with site
  - Website personalization alters site based on who is viewing it
- Customization: ability to change the product to better fit the needs of the customer
  - E-commerce customization focuses on generating personalized product recommendations
- Cookies a basic method to achieve personalization
- Other more sophisticated tools available

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# The Information Policy Set

- Privacy policy
  - Set of public statements declaring how site will treat customers' personal information that is gathered by site
- Accessibility rules
  - Set of design objectives that ensure users with disabilities can effectively access site

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# Insight on Society: Designing for Accessibility

- Class Discussion
  - Why might some merchants be reluctant to make their websites accessible to users with disabilities?
  - How can websites be made more accessible?
  - Should all websites be required by law to provide “equivalent alternatives” for visual and sound content?
  - What additional accessibility problems do mobile devices pose?

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# Developing a Mobile Website and Building Mobile Applications

- Types of m-commerce platforms
  - Mobile website
  - Native app
  - Mobile web app
  - Hybrid app
    - Runs inside native container
    - App distribution
    - Based on HTML5, CSS, JavaScript

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# Planning and Building a Mobile Presence

- Identify business objectives, system functionality, and information requirements
- Choices
  - Mobile website or mobile web app
    - Less expensive
  - Native app
    - Can use device hardware, available offline

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# Table 4.13 Unique Features That Must Be Taken Into Account When Designing a Mobile Presence

| Feature | Implications For Mobile Platform |
|---|---|
| Hardware | Mobile hardware is smaller, and there are more resource constraints in data storage and processing power. |
| Connectivity | The mobile platform is constrained by slower connection speeds than desktop websites. |
| Displays | Mobile displays are much smaller and require simplification. Some screens are not good in sunlight. |
| Interface | Touch-screen technology introduces new interaction routines different from the traditional mouse and keyboard. The mobile platform is not a good data entry tool but can be a good navigational tool. |

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# Mobile Presence Design Considerations

- Platform constraints
  - Graphics, file sizes
- Mobile first design
  - Desktop website design after mobile design
- Responsive web design (RWD)
  - CSS site adjusts layout of site according to device screen resolutions
- Adaptive web design (AWD)
  - Server delivers different templates or versions of site optimized for device

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# Cross-Platform Mobile App Development Tools

- Objective C, Java
- Low cost, open-source alternatives
  - Flutter
  - React Native
  - Appery.io
  - Codiqa
  - Swiftic
  - PhoneGap
  - Axway Appcelerator

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# Mobile Presence: Performance and Cost Considerations

- Mobile first design
  - Most efficient
- Mobile website
  - Resizing existing website for mobile access is least expensive
- Mobile web app
  - Can utilize browser API
- Native app
  - Most expensive; requires more programming

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# Insight on Technology: Klook Sets Its Sights on New Vistas

- Class Discussion
  - What influenced Klook in deciding to create a mobile app?
  - Are there any disadvantages in making a mobile app such a central part of a customer's travel experience?
  - How will the Klook aim to ensure that its expansion into Europe is a success?

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# Careers in E-commerce

- Position: UX Designer
- Qualification/Skills
- Preparing for the Interview
- Possible Interview Questions

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